



DPP-2020-West Phase-3 System Impact Study Report

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MISO

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Executive Summary

This report presents the results of a System Impact Study performed to evaluate the interconnection of the generators in the Definitive Planning Phase (DPP) 2020 West Phase 3. The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models, and results. The ad hoc study group consisted of representatives from the interconnection customers and the utility companies outlined in Table 1 and Table 2.

Table 1: MISO Transmission Owners

MISO Transmission Owner
Dairyland Power Cooperative
ITC Midwest
MidAmerican Energy Company
Northern States Power (Xcel Energy)
Otter Tail Power Company
Southern Minnesota Municipal Power Agency

Table 2: Affected Systems

Affected System
Associated Electric Cooperative Inc.
Manitoba Hydro
Minnkota Power Cooperative Inc.
Southwest Power Pool

24 interconnection requests were studied with a combined nameplate rating of 3,030.1 MW for Energy Resource (ER) and 2,326.1 MW of Network Resource (NR) service. Non-CEI information such as fuel type, project location, and Point of Interconnection may be obtained through the [MISO Public Queue](#). Full project details and modeling assumptions used in this analysis are provided in Appendix M.



Table 3: Interconnection Requests Under Study

Project	Point of Interconnection	Fuel Type	Fuel Breakout	ERIS	NRIS	ERIS Thermal	ERIS Voltage	NRIS	Short Circuit	Stability	AECI	MPC	Total Network Upgrade Cost
J1503	Huntley 345 kV	Solar	Photovoltaic	150	120			\$659,295.00			\$521,699.00		\$1,180,994.00
J1504	Huntley 345 kV	Solar	Photovoltaic	150	120			\$659,295.00			\$521,700.00		\$1,180,994.00
J1520	Ledyard - Colby 345 kV	Solar	Photovoltaic	135	108			\$534,954.00			\$454,469.00		\$989,423.00
J1528	EAU Claire - AS King 345 kV	Battery Storage	Battery Storage	0	0		\$729,750.43						\$729,750.43
J1529	Carroll - Auburn 69 kV line	Solar	Photovoltaic	50	50			\$5,000,000.00					\$5,000,000.00
J1530	Hastings - Macedonia 69 kV	Solar	Photovoltaic	50	50			\$59,850.37			\$1,141,693.00		\$1,201,6435.37
J1532	Clarinda - Coburg 69 kV	Solar	Photovoltaic	50	50			\$140,149.63			\$1,510,795.00		\$1,650,650.63
J1534	Byron 345 kV	Solar	Photovoltaic	100	100	\$24,900,000.00		\$724,564.00					\$25,624,564.00
J1566	Rutland 161 kV	Solar	Photovoltaic	150	150			\$822,054.00			\$518,032.00		\$1,340,086.00
J1572	Huntley - Blue Earth 161 kV	Hybrid	Solar/Battery	150	150		\$112,709.69	\$841,822.00			\$279,753.00		\$1,464,013.69
J1575	Warsaw - Donaldson 115 kV	Wind	Wind Turbine	70	70	\$600,000.00	\$132,111.52	\$498,602.00				\$558,515.00	\$1,789,228.52
J1581	Nobles County 345 kV	Solar	Photovoltaic	200	200			\$1,044,297.00			\$1,162,247.00		\$2,206,544.00
J1588	Bison 345 kV	Solar	Photovoltaic	200	200			\$1,364,031.00				\$1,641,485.00	\$3,005,516.00
J1590	O'Brien - Raun 345 kV	Wind	Wind Turbine	200	40		\$143,991.40	\$86,749.00			\$1,241,109.00		\$1,471,849.40
J1605	Sherburne 34kV	Solar	Photovoltaic	200	200			\$1,537,529.00					\$1,537,529.00
J1611	Langola 115 kV	Solar	Photovoltaic	120	120			\$903,846.00					\$903,546.00
J1620	Pipestone - Split Rock 115 kV	Solar	Photovoltaic	125	0						\$443,547.00		\$443,547.00
J1657	Pine Lake - Apple River 161 kV	Solar	Photovoltaic	100	0								
J1730	EAU Claire - AS King 345 kV	Solar	Photovoltaic	300	300		\$785,134.25	\$2,544,648.00			\$1,316,370.00		\$4,646,152.25
J1734	Wever 161 kV	Solar	Photovoltaic	150	150	\$55,000.00		\$1,318,365.55					\$1,373,365.55
J1769	J898 POI 161 kV	Wind	Wind Turbine	80	0		\$90,919.95						\$90,919.95
J1808	Barron 161 kV	Solar	Photovoltaic	148	0								
J1826	Cannon Falls 69 kV	Hydro	Hydroelectric	4	0		\$5,382.76						\$5,382.76



The full list of network upgrades required are detailed in Network Upgrade Cost Allocation Summary. The cost responsibility for these upgrades has been allocated in accordance with MISO BPM-015. Please refer to Appendix A (CEII) for the cost breakdown for each Interconnection Request and the associated milestone payments required to advance to the next phase.

Model Development

The origin of the 2020 models is the MTEP 20 Target A models with the Bench Cases including all prior-queued projects, while the Study Cases contain all the interconnection requests in 2019, in addition to all the facilities in the Bench Cases.

Model Development and Study Assumptions

Base Case Models

The origin of the DPP 2020 Phase III models is the MTEP20 models with the Bench Cases including all pre-queued projects and associated network upgrades, while the Study Cases contain all the interconnection requests in DPP 2020 Phase III, in addition to all the facilities in the Bench Cases. Below is the list of different scenarios.

- Bench Cases
 - DPP21-2026SH-Phasel-Bench
 - DPP21-2026SUM-Phasel-Bench
- Study Cases
 - DPP21-2026SH-Phasel-Discharging
 - DPP21-2026SUM-Phasel-Discharging
 - DPP21-2026SH-Phasel-Study_Charging

Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the <DPP Study Cycle Phase I/II/III> generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 34.6.1 and TARA version 2001_2_0_0.

Performance Criteria



A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as Injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner FERC filed Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints to obtain any type of Interconnection Service. Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the Deliverability algorithm, to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 "Reactive Power Requirements for Non-Synchronous Generation", which was issued June 16, 2016, states that "Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation." As such, this Final Rule applies to all wind and solar projects included in this study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter-based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

Results of the FERC Order 827 Compliance Review are detailed in Appendix N (CEII).

Steady State Thermal Analyses

The steady state ERIS and NRIS (deliverability) analyses are conducted to identify what network upgrades, if any, are required to reliably interconnect current study interconnection requests to the MISO transmission system. Section 6 of the MISO BPM-015 details the processes and methodologies used for the DPP steady state thermal analysis.

ERIS Analysis

The following models were developed for the ERIS analysis:



- Summer Shoulder Bench
- Summer Shoulder Discharging Study
- Summer Shoulder Charging Study
- Summer Peak Bench
- Summer Peak Discharging

The origin of the DPP Study Cycle Phase III models is the MTEP20 models with the Bench Cases including all pre-queued projects and associated Network Upgrades, while the Study Cases contain all the interconnection requests in DPP Study Cycle Phase III, in addition to all the facilities in the Bench Cases

MISO utilized PSS®E 34.6.1 and PowerGEM TARA v2202.2.0.0 for the ER analysis. AC Contingency calculations were performed on the bench and study models and identified network constraints in accordance with the MISO BPM-015. The full list of constraints identified in the ERIS and NRIS analysis are detailed in Appendix C (CEII) and Appendix D (CEII).

Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

Contingencies

The following contingencies were considered in the steady state analysis:

1. NERC Category P0 (system intact – no contingencies)
2. NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
3. NERC Category P2-P7 contingencies
4. For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

Solutions to constraints observed in the system impact study were developed in partnership with the host and affected transmission owners. Network upgrades were cost allocated in accordance with the MISO BPM-015. The list of network upgrades required for interconnection service are detailed in Network Upgrade Cost Allocation Summary. Cost allocation of those upgrades along with the required milestone payments to advance to the next phase are outlined in Appendix A (CEII).

Network upgrade cost estimates were developed on a planning level and are later refined in the Phase II facility studies. Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator. Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.

NRIS (Deliverability) Analysis

Generator interconnection projects must pass the Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource project or to proceed with the system upgrades that will make the generator fully deliverable. The Generator Deliverability Study ensures that the Network Resources, on an aggregate basis,



can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind and solar generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

The MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

The model developed for the NRIS analysis was derived from the Summer Peak Discharging case from the ERIS analysis; however, the study units are not dispatched as this occurs through the TARA deliverability analysis.

Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all its output is not deliverable. The restricted MW is calculated as following:

$$(MW\ restricted) = \frac{worst\ loading - MW\ rating}{(generator\ sensitivity\ factor)}$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

Shared Network Upgrades Analysis

The Shared Network Upgrade (SNU) Analysis, which tests for Network Upgrades driven by higher queued interconnection projects, was performed for this System Impact Study. Please refer to Appendix A for maximum MW impacts and SNU cost allocations.

Stability Analysis

Stability analysis is performed on the study units to determine their dynamics performance. Each region performs stability analysis using the dispatch from the ERIS discharging case. Supplemental analysis may also be performed to capture relevant Local Planning Criteria (LPC) for new generator interconnections.

Model Development

Stability models are developed using the ERIS five year out summer peak model. LPC stability models are developed based on the stability study models and adjusted to comply with the relevant LPC by fully dispatching nearby local generation.

- Bench Case
 - DPP20-2025SH90-PhaseIII-Bench.pfb
 - DPP20-2025SUM-PhaseIII-Bench.pfb
- Study Case
 - DPP20-2025SH90-PhaseIII-Study_Discharging.pfb
 - DPP20-2025SH90-PhaseIII-Study_Charging.pfb
 - DPP20-2025SUM-PhaseIII-Study_Discharging.pfb

Study Methodology

The purpose of the study is to identify potential angular instabilities, voltage dip violations, and damping violations, if any, due to the interconnection of the projects in the study cycle under disturbance conditions, and the impact of all study projects on the system stability performance.

The fault scenarios simulated in this study cover faults simulated as part of the MTEP 20 analysis as well as selected three-phase (3PH) faults with normal clearing and single line to ground (SLG) faults with delayed



clearing. Dynamic simulations of fault scenarios were performed using the DSATools TSAT program (version 20.0).

Fault scenarios were first simulated using the study case and the results were reviewed. For scenarios that exhibited instability, the bench case was simulated such that the stability performance with and without the proposed interconnection projects could be compared. Any new stability problems attributed to the proposed interconnection projects are flagged and reported.

For each fault, rotor angles, speed deviation, and electrical power outputs of the study generators and the generators in the proximity were monitored. Voltages at selected buses, including all POI buses of the study projects and all future buses, were also monitored.

In addition, all renewable study projects are subject to the voltage ride-through and frequency ride-through criteria specified in NERC PRC-024-2 ("Generator Frequency and Voltage Protective Relay Settings"¹) to check if the projects remain connected during frequency and voltage excursions. Specifically, PRC-024-2 mandates that protective relaying should be set in such a way that:

- Voltage Ride-Through: a generator shall withstand zero voltage at the POI (typically the primary side of the station transformer) for up to 0.15 seconds (9 cycles) and the ensuing voltage recovery period for three phase faults.
- Frequency Ride-Through: a generator shall maintain continuous operation between 59.5 and 60.5 Hz.

Local Planning Criteria

Otter Tail Power Company

Section 3.1 of the "OTP TO Planning Criteria" states the following:

Interconnections to the Otter Tail transmission system must be evaluated to ensure that a reliable transmission system can be maintained after a proposed interconnection is constructed. In the event that a new interconnection identifies a reliability concern through studies of the transmission system, it will be necessary to identify and Otter Tail Planning Criteria, Guidelines and Study Procedures Page 14 implement necessary system upgrades to ensure acceptable system performance is achieved with the new interconnection. Interconnections to the Otter Tail system should follow the Guidelines for Generation, Tie-Lines, and Substation Interconnections, which can be found at the following link:

<https://www.otpc.com/media/1243/otptechguidecasot.pdf>

New interconnections should be studied using a variety of models and scenarios to ensure that the system maintains reliability under a variety of conditions.

Study Results

The complete list of 3PH and SLG faults simulated as well as their corresponding results and plots can be found in Appendix F.

Dairyland Power Cooperative

Section 1.1 of the "DPC TO Planning Criteria" states the following:

For generation interconnection studies all existing local generation must be analyzed at full output at both the minimum and peak load conditions. The new generation will be added on top of that local area generation to determine if the total generation in the area can be delivered to the MISO load on the transmission system.

Study Results

¹ <http://www.nerc.com/pa/Stand/Reliability%20Standards/PRC-024-2.pdf>



The complete list of 3PH and SLG faults simulated as well as their corresponding results and plots can be found in Appendix F.

Short Circuit Analysis

Short circuit analysis was performed utilizing CAPE software. Single line to ground faults and three phase faults were evaluated for pre- and post-project cases (comparable to Bench and Study models). The details pertaining to the short circuit analysis are presented in Appendix G (CEII) – Short Circuit Analysis.

Affected System Impact Study

The details pertaining to the Affected Systems studies are in Appendix G (CEII)

Short Circuit Analysis

Appendix H-K (CEII)

Affected Systems Reports.



Appendices

Appendix A (CEII)

- Interconnection Cost Summary
- Milestone Payment Summary
- Shared Network Upgrade Summary

Appendix B (CEII)

- Network Upgrade Cost Allocation Summary

Appendix C (CEII)

- ERIS Thermal Analysis
- ERIS Thermal Cost Sharing
- ERIS Voltage Analysis
- ERIS Voltage Cost Sharing

Appendix D (CEII)

- NRIS Results
- NRIS Cost Allocation

Appendix E (CEII)

- Stability Analysis

Appendix F (CEII)

- LPC Analysis

Appendix G (CEII)

- Short Circuit Analysis

Appendix H-K (CEII)

- Affected Systems Reports

Appendix L (CEII)

- A10 Results

Appendix M (CEII)

- Project Data

Appendix N (CEII)

- FERC 827 Analysis